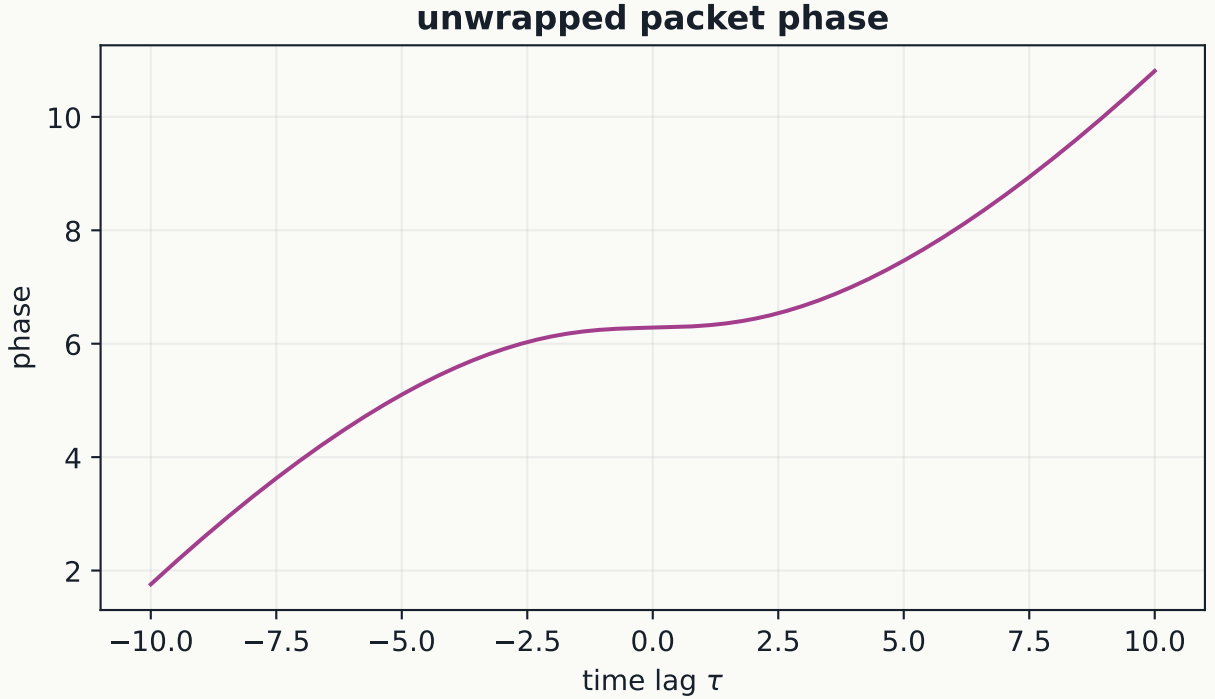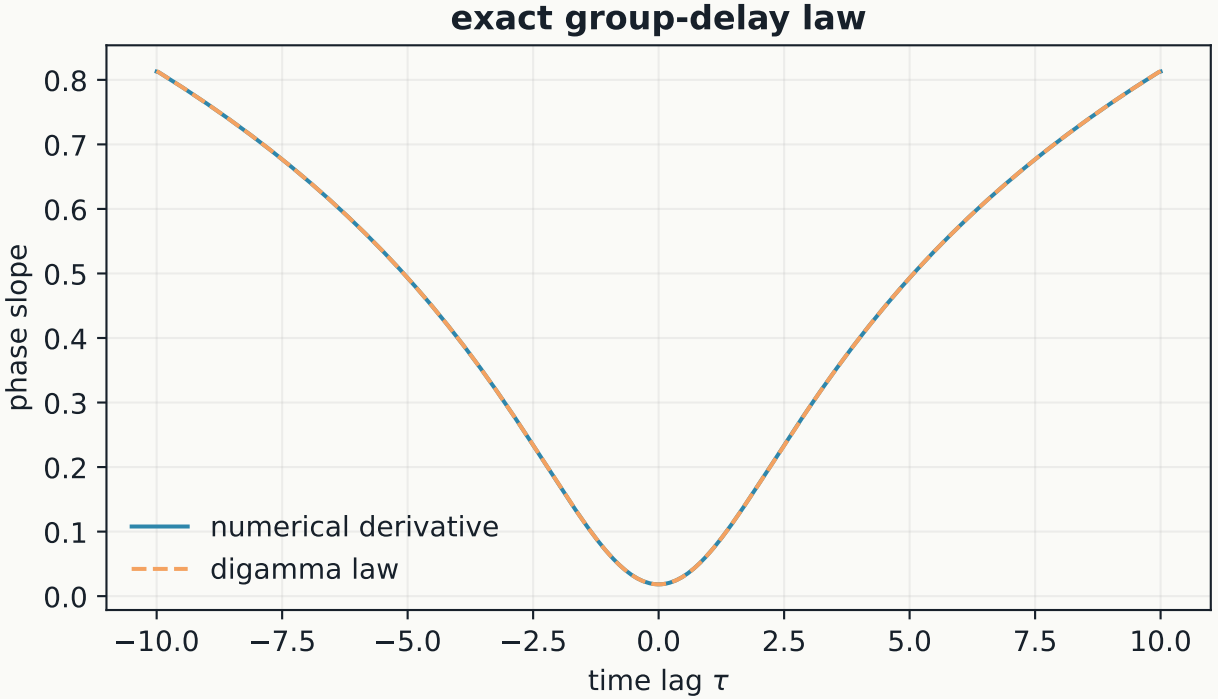The limiting phase is controlled by the digamma function